Task 03: Leveraging Nmap Scripting Engine (NSE)

What the Nmap Scripting Engine (NSE) Is:

The Nmap Scripting Engine (NSE) is a feature in Nmap that allows you to run small scripts written in Lua to automate tasks such as service discovery, vulnerability detection, and information gathering.

Instead of manually checking every service, NSE scripts automatically test services for common weaknesses and configurations.

Running Discovery and Safe Category Scripts

COMMAND

Nmap -sV –script discovery.safe hackthissite.org

Running DiscoveryScripts

➔ **-sV**

This flag enables service version detection so Nmap can identify the exact services and versions running on open ports.

➔ ***--script discovery***

This option runs scripts from the Discovery category that gather information about the target, such as network services, hostnames, and system details.

➔ ***--script safe***

This option runs scripts that are considered safe and non-intrusive, meaning they collect useful information without affecting or damaging the target system.

POC

```
└─$ nmap -sV --script discovery,safe hackthissite.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:46 +0500
No profinet devices in the subnet
Pre-scan script results:
| broadcast-pim-discovery: ERROR: Script execution failed (use -d to debug)
| broadcast-igmp-discovery: ERROR: Script execution failed (use -d to debug)
| broadcast-pppoe-discover: ERROR: Script execution failed (use -d to debug)
| http-robtex-shared-ns: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| broadcast-eigrp-discovery: ERROR: Script execution failed (use -d to debug)
| mrinfo: ERROR: Script execution failed (use -d to debug)
| targets-ipv6-multicast-echo:
|   IP: fe80::1                    MAC: 14:ca:56:9f:ff:6b  IFACE: wlan0
|   IP: fe80::50bd:7aff:fe05:7cf5  MAC: 52:bd:7a:05:7c:f5  IFACE: wlan0
|_  Use --script-args=newtargets to add the results as targets
| targets-asn:
|_  targets-asn.asn is a mandatory parameter
| targets-sniffer: Sniffed 0 address(es).
| broadcast-ospf2-discover: ERROR: Script execution failed (use -d to debug)
Bug in broadcast-upnp-info: no string output.
| broadcast-dhcp-discover:
|   Response 1 of 1:
|     Interface: wlan0
|     IP Offered: 192.168.1.12
|     Subnet Mask: 255.255.255.0
|     Router: 192.168.1.1
|     Domain Name Server: 192.168.1.1
|_    Server Identifier: 192.168.1.1
| broadcast-listener:
|   udp
|       DHCP
|         srv ip      cli ip        mask          gw            dns           vendor
|           192.168.1.1  192.168.1.11  255.255.255.0  192.168.1.1  192.168.1.1  -
|_          192.168.1.1  192.168.1.12  255.255.255.0  192.168.1.1  192.168.1.1  -
Bug in lltd-discovery: no string output.
| knx-gateway-discover: ERROR: Script execution failed (use -d to debug)
| eap-info:
| Available authentication methods with identity="anonymous" on interface eth0
|   unknown  EAP-TTLS
|   unknown  EAP-TLS
|   unknown  EAP-MSCHAP-V2
|_  unknown  PEAP
| hostmap-robtex: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| targets-ipv6-multicast-invalid-dst:
|   IP: fe80::1  MAC: 14:ca:56:9f:ff:6b  IFACE: wlan0
|_  Use --script-args=newtargets to add the results as targets
```

Discovery Category All Scripts

The **Discovery category** contains scripts that help identify hosts, services, and network information.

http-enum
This script enumerates common directories and files on a web server to discover hidden resources like admin panels or login pages.

dns-brute
This script performs DNS brute-forcing to discover subdomains associated with the target domain.

snmp-info
This script retrieves information from SNMP-enabled devices such as system name, location, and contact information.

broadcast-dhcp-discover
This script sends DHCP requests on the network to discover DHCP servers and network configuration details.

smtp-commands
This script lists the commands supported by an SMTP mail server, helping identify its capabilities and configuration.

ftp-anon
This script checks whether anonymous login is allowed on an FTP server.

Safe Category Script

The **Safe category** includes scripts that gather information without performing aggressive or intrusive actions.

ssl-cert
This script retrieves SSL/TLS certificate details such as issuer, validity period, and encryption information.

http-title
This script retrieves the title of a webpage, which can help identify the type of web application or service running.

http-headers
This script displays HTTP response headers from a web server to reveal server information and configurations.

ssh-hostkey
This script retrieves the SSH public host key from an SSH server.

Banner
This script attempts to read the service banner from open ports to identify software versions or server information.

whois-ip
This script performs a WHOIS lookup to gather information about the ownership of an IP address.

Running Specific Scripts, Together You can run several individual

scripts at once.

nmap -p 80,443 --script http-enum,ssl-cert,http-title hackthissite.com

Explanation -p 80,443

This option scans specific ports (HTTP and HTTPS).

--script http-enum,ssl-cert,http-title This tells Nmap to run multiple scripts at the same time.

vuln

These scripts check whether the target system is vulnerable to known security vulnerabilities and often map results to CVE IDs.
command:

nmap -sV --script vuln hackthissite.com

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -T4 --script vuln hackthissite.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 17:00 +0500
Nmap scan report for hackthissite.com (76.223.54.146)
Host is up (0.057s latency).
Other addresses for hackthissite.com (not scanned): 13.248.169.48
rDNS record for 76.223.54.146: a904c694c05102f30.awsglobalaccelerator.com
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE        VERSION
80/tcp   open  http-proxy     HAProxy http proxy 2.0.0 or later
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-majordomo2-dir-traversal: ERROR: Script execution failed (use -d to debug)
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
443/tcp open  ssl/tcpwrapped
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
|_http-majordomo2-dir-traversal: ERROR: Script execution failed (use -d to debug)
Service Info: Device: load balancer

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 36.40 seconds

┌──(hyena㉿hyena)-[~/Downloads]
```